1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L11 – Introduction to Design (Interface & Code)**

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4 **Recap**

- We have covered usability requirement gathering, analysis and specification
  - Contextual inquiry for gathering
  - Affinity diagram for analysis
  - SRS format for specification

5 **Today's Topic**

- We shall start discussion on the important life cycle stage of "design"

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7 **Design Concerns**

8 **Design**

- Two issues
  - Where to start?
  - How to represent (design language)?

9 **Where to Start?**

- Creative thinking!
  - May be aided by intuition
  - May be aided by experience/domain knowledge

10 **Where to Start?**

- In UCD (user-centered design), we have TWO concerns
  - Interface design
  - Code design

11 **Interface Design**

12 **Designing the Interface**

- In interactive system, key element is user-system interface (and also interaction between them)
- Usability primarily depends on both

13 **Where to Start?**

- Prime application of "creativity"
- Can rely on experience
- Can rely on intuition

14 **Where to Start?**

- Can also make use of some guidelines/thumb rules/checklists/heuristics as starting point (to aid our creative thinking)

15 **Design Guidelines**

- Many guidelines are there

16 **Design Guidelines**

- Some very generic and consequently small in size - covers broad aspects of interactive systems, at a rather high level
  - "eight golden rules" (Shneiderman, 1986)
  - "seven principles" (Norman, 1988)

17 **Design Guidelines**

- Others are more detailed and specific and therefore large in (set) size - intended to cover minute aspects of the design, often for specific products
  - "human interface guidelines" for the Apple systems

18 **Where to Start?**

- Shall learn about two such set of guidelines
  - 8 golden rules of Shneiderman
  - Norma's 7 principles

19 **Note**

- The design-prototype-evaluate cycle primarily caters to interface design
- Once interface design stabilized and no further iteration required, we focus on code design

20 **Code Design**

21 **Code Design**

- Altogether different challenges
- Main idea is to make the code "manageable"

22 **Code Design**

- Done based on SRS
- Two phases
  - Preliminary (high-level) design
  - Detailed design (also called module-specification document)

23 **High-Level Design**

- Identification of modules
- Control relationships between modules
- Definition of interfaces between modules

24 **Detailed Design**

- Identification of data structures and algorithms for different modules

25 **Design Language**

26 **Design Language**

- Refers to a specification of design
- Different "languages" for interface and code design

27 **Interface Design Language**

- To express interface and interaction design ideas, we can take recourse to "prototypes"
  - Can be considered the language for interface design
  - We shall learn about prototyping in details in a later lecture

28 **Code Design Language**

- Code design can be expressed in various ways
  - Natural languages
  - Semi-formal languages
  - Formal languages

29 **Code Design Language**

- We shall learn about two semi-formal (graphical languages)
  - DFD
  - UML

30 **Note**

- In the next lecture, we shall learn about design guidelines